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BELLOWS-L2: Hardware LZ4 Compression and Decompression

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 29 October 2026 (2026-10-29 21:29 UTC+8 Beijing Time)

Abstract

BELLOWS-L2 combines a block-buffered hardware LZ4 compressor with the preserved BELLOWS-L1 streaming decoder. Byte-wide interfaces support independent simultaneous compression and decompression or selectable internal compress-then-decompress operation. Registered four-byte hash lookup, literal-anchor buffering and a token emitter implement actual match compression. Software-vs-Hardware numerical comparison passed all 18 capacity/seed configurations with byte-exact recovery. Each of six required repetitive fixtures achieved compression ratio below 1.0: 4096-byte inputs produced 26–282 encoded bytes, and 65536-byte inputs produced 267–523 encoded bytes. These results establish agreement for executed cases, not exhaustive mathematical proof. [E1187, E1641] Native L1 RTL timing lint, designated L1.5 in the work order, passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA reported WNS_setup = + 3.405 ns at a 10.0 ns clock period on Arria 10 GX 10AX115N2F45E1SG. This standalone-IP configuration uses virtual pins except for clk and only a clock constraint, establishing internal timing closure rather than board-level interface timing. L3 recorded three consecutive deterministic physical FPGA HIL passes; retained HIL output documents register diagnostics, while portable simulations support the payload-level numerical comparisons. [E1642, E1643, E1644]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The single-clock design retains the three parent decoder modules byte-for-byte and adds a separate compressor hierarchy. Compression collects a complete bounded block before searching and encoding. Buffering permits overlap-safe comparison against original bytes and exact literal-only fallback before encoded output becomes externally visible. Synchronous memories and serial comparison trade throughput and latency for bounded combinational depth. Blocks below are arranged row-major in three columns. [E001–E010]

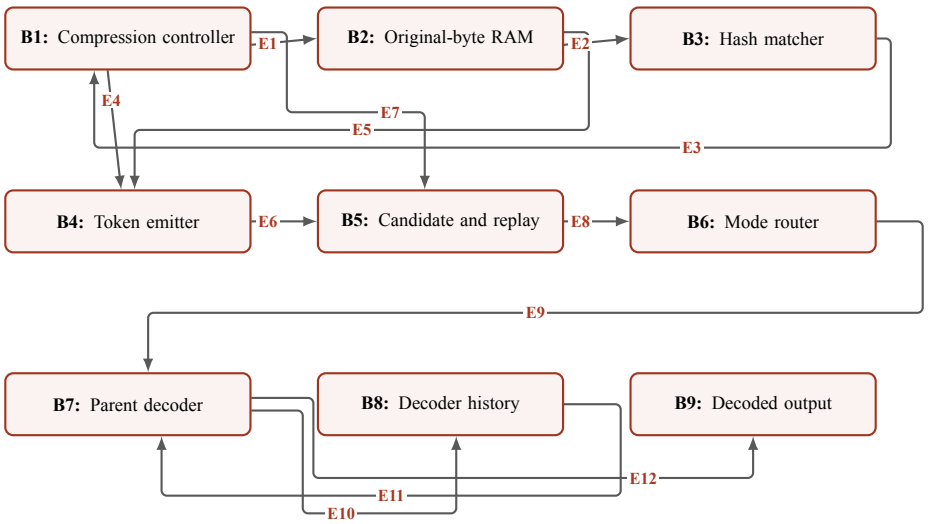


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Compression controller	bellows_l2_compressor owns collection, framing errors, discard, block length, literal anchor, memory-port arbitration and completion status.
B2: Original-byte RAM	bellows_l2_literal_buffer stores MAX_BLOCK_BYTES bytes with synchronous reads: 65536 x 8 logical bits at default capacity.
B3: Hash matcher	bellows_l2_hash_match acquires four-byte keys, registers hash lookup, validates candidates and extends matches serially. Storage includes 256 positions of 16 bits and 256 validity bits.
B4: Token emitter	bellows_l2_emitter accepts sequence descriptors and registers tokens, extensions, literals and little-endian offsets.
B5: Candidate and replay	Controller-owned synchronous RAM stores 2*MAX_BLOCK_BYTES+16 encoded bytes, or 131088 bytes by default. Completed-size comparison selects candidate replay or literal-only regeneration.
B6: Mode router	bellows_l2_top directs compressed data externally or into the decoder. Independent mode permits both lanes to operate concurrently.
B7: Parent decoder	Unmodified bellows_l1_top parses raw LZ4 and reconstructs bytes while retaining parent framing, error and status behavior.
B8: Decoder history	Unmodified bellows_l1_history supplies synchronous 65536-byte rolling storage. History ownership is invalidated between blocks.
B9: Decoded output	Unmodified bellows_l1_output stages pending and presented bytes, identifies the final byte and preserves held output under backpressure.

Directed relationship	Meaning
E1: B1 to B2	Accepted raw-byte writes
E2: B2 to B3	Registered search bytes
E3: B3 to B1	Match descriptors and done
E4: B1 to B4	Literal-span and match descriptors
E5: B2 to B4	Registered literal bytes
E6: B4 to B5	Encoded sequence bytes
E7: B1 to B5	Size selection and replay control
E8: B5 to B6	Compressed ready/valid stream
E9: B6 to B7	Selected external or internal stream
E10: B7 to B8	History writes and read requests
E11: B8 to B7	Registered history response
E12: B7 to B9	Decoded candidates and finish

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk; rst_n; roundtrip_ enable	Input	1 bit each	Rising-edge clock, asynchronous active-low reset and route selection. Change mode only during reset or when both lanes have no active block, output or outstanding status.

Signal / group	Direction	Width	Contract
c_s_data; c_s_keep; c_s_valid; c_s_last; c_s_ready	Inputs: data, keep, valid, last; output: ready	Data 8 bits; others 1 bit	Raw compressor ingress. Normal bytes require keep=1. A sole keep=0,last=1 beat represents an empty block, with ignored data.
c_m_data; c_m_valid; c_m_last; c_m_ready	Outputs: data, valid, last; input: ready	Data 8 bits; others 1 bit	Raw-LZ4 compressed output. In internal round-trip mode all three outputs are zero and c_m_ready is ignored.
c_status_valid; c_status_error; c_status_ready	Outputs: valid, error; input: ready	Error 3 bits; others 1 bit	Held compressor completion. Status follows final output retirement or completed error discard; acknowledgement permits the next raw block.
d_s_data; d_s_valid; d_s_last; d_s_ready	Inputs: data, valid, last; output: ready	Data 8 bits; others 1 bit	External raw-LZ4 decoder ingress, without a keep signal. d_s_ready is zero during internal round-trip operation.
d_m_data; d_m_valid; d_m_last; d_m_ready	Outputs: data, valid, last; input: ready	Data 8 bits; others 1 bit	Recovered byte stream. Empty decoded blocks produce no output beat.
d_status_valid; d_status_error; d_status_ready	Outputs: valid, error; input: ready	Error 3 bits; others 1 bit	Preserved decoder completion channel. Both status channels remain independently observable and handshaken in either mode.

Parameter	Default	Meaning / supported values
MAX_BLOCK_BYTES	65536	Integer capacity from 1 through 65536 inclusive, interpreted as an unsigned datapath limit. Values 0 and 65537 receive explicit fatal diagnostics before transactions. Capacities 1, 5, 13, 257, 4096 and 65536 were exercised. This parameter does not limit decoder output length. bellows_l2_compressor exposes the same parameter and compressor interfaces without their c_ prefix.

III Behavior and corner cases

A Transfers, reset and completion

Only pre-edge valid&&ready transfers a beat. Stalled senders hold valid and associated data, last and keep stable. All public outputs, including ready, are zero during reset. Reset abandons transactions and invalidates ownership without clearing large data arrays. Processing begins after collection closes in registered state, never through combinational input forwarding. No next compressor block is accepted until its status is acknowledged.

B Frozen match search

At ascending visited position p, compute h=b0 XOR rol8(b1,1) XOR rol8(b2,2) XOR rol8(b3,3), with eight-bit rotation. Each of 256 direct-mapped entries retains the latest inserted absolute position. Lookup reads the old entry before replacing it with p. Candidate q requires validity, q<p, offset 1..65535 and four equal bytes. Greedy extension permits overlap. Misses advance one byte; matches resume at their end without inserting skipped starts. Hash validity clears for every block.

C End restrictions and encoding

Blocks shorter than 13 bytes are literal-only. Otherwise match starts satisfy p<=N-12 and extensions end at or before N-5. Minimum match length is four, and every block ends with a literal-only sequence. Token nibbles encode min(literal_length,15) and min(match_length-4,15); the final sequence has low nibble zero. Saturation requires repeated 255 extension bytes followed by a remainder 0..254, including a terminating zero when required. Emit literals before the nonzero offset, serialized low byte first, and then any match extension. Empty input encodes as 0x00.

D Literal fallback

Literal-only size is N+1, plus floor((N-15)/255)+1 for N>=15. Retain the completed candidate only when strictly smaller; ties and larger candidates regenerate the exact literal-only encoding. Hardware computes extensions through registered subtraction rather than division. Verification checks the conservative candidate-storage bound.

E Registered stages

Four request/capture pairs acquire a key; separate stages register the hash, synchronous table response and candidate validation. Four further request/capture comparisons establish the minimum match. Extension uses check/request/capture stages. The emitter latches descriptors and registers bytes. Public replay uses request/load/hold states, inserting bubbles even with continuously ready output. Parent literals pass through commit staging; match bytes use synchronous history-read and commit states. Consumer stalls and variable search work prevent a fixed end-to-end cycle count.

F Errors and recovery

Compressor codes are 0 success, 5 capacity overflow and 6 malformed keep; other codes are reserved. First detected error wins, with malformed keep taking precedence when both arise on one beat. Error blocks emit no compressed output and drain through accepted last before status. Decoder codes remain 0 success, 1 truncation, 2 zero offset, 3 unavailable history and 4 length overflow, preserving valid-prefix output and subsequent recovery.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison and lossless recovery

Method. Independent Python raw-LZ4 parser and direct-sequence generator; actual Icarus and Verilator behavior checks. Six capacities crossed with seeds 1,17,89.

Result. 18/18 matrix cells passed. The ledger records 508 independently parsed compressed blocks, 504 hardware-compress/hardware-decompress cases and 80 internal-round-trip blocks. Complete-byte comparisons reported zero mismatches. Coverage counters include 1275086 compressor-reference bytes and 1770235 decoder-reference bytes, including repeated test traffic.

Evidence: E011, E012, E1187, E1641

B Each structured fixture has ratio below 1.0 and real matches

Method. Measure actual RTL encoded lengths and independently parse matches; run each fixture under all three seeds. Ratio is encoded/input bytes, excluding status.

Result. At N=4096: repeated A 26/4096=0.00634765625; ABCD 29/4096=0.007080078125; byte ramp 282/4096=0.06884765625. At N=65536: A 267/65536; ABCD 270/65536; byte ramp 523/65536. All six pass separately, contain matches and retain five final literals.

Evidence: E1187, E1641

C Backpressure, boundaries and recovery

Method. Public-pin assertions, independent ingress/egress/status stalls, simultaneous duplex traffic, resets and directed format/history cases.

Result. Passed held-final and held-status checks on applicable channels, 377 exact fallback comparisons, 152 malformed/post-error-recovery cases, offset-65535 and history-wrap cases, output beyond 65536 bytes, extension boundaries and reset coverage. Illegal capacity endpoints were explicitly rejected.

Evidence: E011, E015, E1187, E1641

D Defect sensitivity

Method. Compile and execute three disposable synthesizable RTL mutants without changing expected results.

Result. Offset corruption reached ZERO_OFFSET rejection; trailing-limit violation reached TRAILING_FIVE_LITERALS rejection; stalled-output corruption reached HELD_C_OUTPUT rejection. Compilation failure or timeout was not counted as detection.

Evidence: E1187, E1641

E Synthesizable hierarchy and bounded reproduction

Method. Real Yosys synthesis at all six tested capacities; a clean declared-input copy runs targeted Icarus duplex/round-trip/fallback traffic and synthesis.

Result. Six synthesis configurations retained all eight required modules, state and memories with zero latch cells. Bounded clean-directory reproduction passed; it was not a second full matrix.

Evidence: E1187, E1641

F L1 RTL timing lint / native L1.5

Method. Native hazard analysis of eight RTL files.

Result. Passed with zero divider/modulo, mixed sequential assignment, LUT clock-gate and AST loop-unroll hazards.

Evidence: E1642

G L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit analysis of bellows_l2_top on Intel Arria 10 GX 10AX115N2F45E1SG, using a standalone-IP configuration. All ports except clk are virtual pins; the supplied timing constraint creates only a 10.000 ns clock.

Result. Passed: WNS_setup = +3.405 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 151.63 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1643

H L3 physical Arria 10 FPGA HIL

Method. Native hardware-bridge gate with three consecutive physical runs.

Result. 3/3 passed with deterministic results and a healthy bridge. Retained output documents register-57 diagnostics, not byte-level compression/decompression traffic or physical Software-vs-Hardware payload parity measurements.

Evidence: E1644

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 5	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

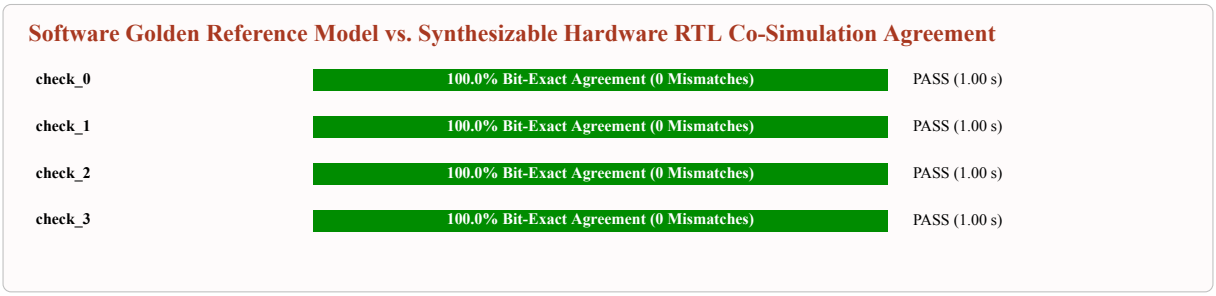


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +3.405 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

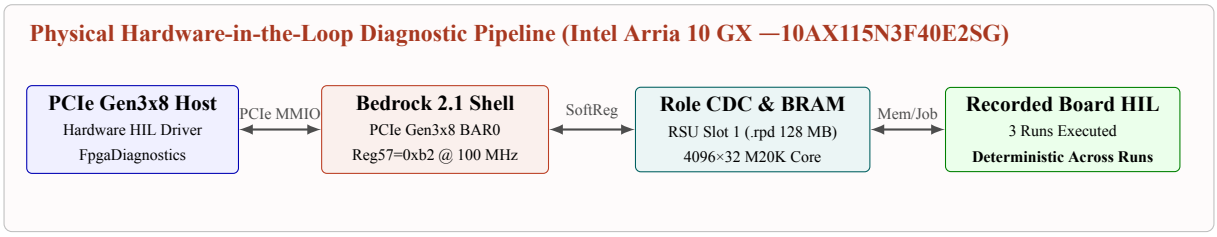


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Instantiate bellows_l2_top from rtl/*.sv. Use design/SPEC.md for the public contract and design/ARCHITECTURE.md for memory ownership and local stages. Compile only deliverable RTL copies, not duplicate parent module definitions from the reference directory.

Assert rst_n low before traffic. Select roundtrip_enable at reset or complete lane quiescence. In duplex mode drive independent c_s_* and d_s_* streams. In round-trip mode drive raw c_s_* traffic and consume d_m_*; acknowledge both status channels.

Hold payloads stable until acceptance. Mark each nonempty raw block's last byte with c_s_last=1 and keep=1. Represent empty input with one keep=0,last=1 beat. Drain and inspect status before beginning the next compression block.

Run python3 tests/verify.py --output evidence/l1 from a source copy where that destination does not exist. Occupied outputs are refused. Dependencies include Python 3, Icarus/vvp, Verilator, a C++ compiler, make and Yosys; no external Python package is required.

The bounded invocation is python3 tests/verify.py --output result --subset reproduce with a fresh destination. The full verifier performs clean-copy reproduction automatically. Numerical records are in evidence/l1/ results.json; command receipts and tool versions accompany the execution bundle.

Keep native Quartus and physical hardware execution separate from portable L1. Final integration needs its own board pin assignments, external timing constraints and applicable PCIe-interface constraints; the reported standalone-IP STA configuration does not supply these.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 8 Verilog/SystemVerilog files; 1094 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/bellows_l1_history.sv	25 total	22 code
rtl/bellows_l1_output.sv	65 total	61 code
rtl/bellows_l1_top.sv	307 total	281 code
rtl/bellows_l2_compressor.sv	282 total	269 code
rtl/bellows_l2_emitter.sv	185 total	183 code
rtl/bellows_l2_hash_match.sv	193 total	186 code
rtl/bellows_l2_literal_buffer.sv	36 total	34 code
rtl/bellows_l2_top.sv	61 total	58 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E011	tests/bench.py
E012	tests/oracle.py
E015	tests/verify.py
E1187	.vulcan/engineering/run-0001/attempt-0621/stdout.log
E1641	.vulcan/engineering/run-0001/attempt-0910/response.json
E1642	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1643	.vulcan/engineering/run-0001/quartus-sta.json
E1644	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).